

Performance Grading

Performance Grades

Max. Design Temp.	PG 46	PG 52						PG 58						PG 64						PG 70						PG 76						PG 82					
Min. Design Temp.	-34	-40	-46	-10	-16	-22	-28	-34	-40	-46	-16	-22	-28	-34	-40	-10	-16	-22	-28	-34	-40	-10	-16	-22	-28	-34	-40	-10	-16	-22	-28	-34	-10	-16	-22	-28	-34

Original

≥230 °C	Flash Point						
≤ 3 Pa-s @ 135 °C	Rotational Viscosity						
≥ 1.00 kPa	DSR G*/sin δ (Dynamic Shear Rheometer)						
	46	52	58	64	70	76	82

(Rolling Thin Film Oven) RTFO, Mass Change $\leq 1.00\%$

≥ 2.20 kPa	DSR $G^*/\sin \delta$ (Dynamic Shear Rheometer)						
	46	52	58	64	70	76	82

(Pressure Aging Vessel) PAV

20 hours, 2.10 MPa	90			90			100			100			100(110)			100(110)			100(110)																			
≤ 5000 kPa	DSR $G^*\sin \delta$ (Dynamic Shear Rheometer)																												Intermediate Temp. = [(Max. + Min.)/2] + 4									
	10	7	4	25	22	19	16	13	10	7	25	22	19	16	13	31	28	25	22	19	16	34	31	28	25	22	19	37	34	31	28	25	40	37	34	31	28	
$S \leq 300$ MPa $m \geq 0.300$	BBR S (creep stiffness) & m-value (Bending Beam Rheometer)																																					
	-24	-30	-36	0	-6	-12	-18	-24	-30	-36	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	0	-6	-12	-18	-24	

If BBR m -value ≥ 0.300 and creep stiffness is between 300 and 600, the Direct Tension failure strain requirement can be used in lieu of the creep stiffness requirement.

$\epsilon_f \geq 1.00\%$	DTT (Direct Tension Tester)																																			
	-24	-30	-36	0	-6	-12	-18	-24	-30	-36	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	-30	0	-6	-12	-18	-24	0	-6	-12	-18